

fair_gbm

FairGBMConfig

```
FairGBMConfig(constraint\_group\_column: str = '', constraint\_type: Optional[str] = 'FPR', global\_constraint\_type: Optional[str] = 'FPR,FNR', global\_target\_fpr: Optional[float] = 0.05, global\_target\_fnr: Optional[float] = 0.5, objective: Optional[str] = 'constrained_cross_entropy', constraint\_fpr\_threshold: Optional[float] = 0.0, constraint\_fnr\_threshold: Optional[float] = 0.0, multiplier\_learning\_rate: Optional[float] = 1000.0, init\_multipliers: Optional[str] = '', constraint\_stepwise\_proxy: Optional[str] = 'cross_entropy', objective\_stepwise\_proxy: Optional[str] = '', **kwargs)
```

Bases: [GenericLightGBMConfig](#)

The FairGBM (LightGBM with fairness constraints) configuration object used for model training.

Parameters:

Name	Type	Description	Default
constraint_group_column	str	(Fairness) Sensitive group column. Must be categorical.	' '
constraint_type	str	(Fairness) Constraint type. Either "FPR", "FNR", "FPR,FNR".	' FPR '
global_constraint_type	str	(Fairness) Global constraint type. Either "FPR", "FNR", "FPR,FNR".	' FPR, FNR '
global_target_fpr	float	(Fairness) Global target FPR.	0.05
global_target_fnr	float	(Fairness) Global target FNR.	0.5
objective	str	(Fairness) Objective function. Either "constrained_cross_entropy", "constrained_recall_objective".	' constrained_cross_entropy '
constraint_fpr_threshold	float	(Fairness) FPR slack for fairness.	0.0
constraint_fnr_threshold	float	(Fairness) FNR slack for fairness.	0.0
	float	(Fairness) Multipliers' learning rate.	1000.0

Name	Type	Description	Default
<code>multiplier_learning_rate</code> ¶			
<code>init_multipliers</code> ¶	str	(Fairness) Initial multipliers.	<code>''</code>
<code>constraint_stepwise_proxy</code> ¶	str	(Fairness) Stepwise proxy for fairness constraints. Either "cross_entropy", "quadratic", "hinge".	<code>'cross_entropy'</code>
<code>objective_stepwise_proxy</code> ¶	str	(Fairness) Stepwise proxy for global constraints. Either "cross_entropy", "quadratic", "hinge".	<code>''</code>

Other Parameters:

Name	Type	Description
<code>boosting_type</code>	str	Boosting type.
<code>num_iterations</code>	int	Number of booster iterations.
<code>learning_rate</code>	float	Learning rate.
<code>num_leaves</code>	int	Maximum tree leaves.
<code>lightgbm_seed</code>	int	Seed used to generate other seeds.
<code>max_depth</code>	int	Maximum tree depth.
<code>min_data_in_leaf</code>	int	Minimum leaf data points.
<code>min_sum_hessian_in_leaf</code>	float	Minimal sum of hessian per leaf.
<code>bagging_fraction</code>	float	Bagging sample fraction.
<code>bagging_freq</code>	int	Bagging frequency.
<code>feature_fraction</code>	float	Feature fraction by tree.
<code>feature_fraction_by_node</code>	float	Feature fraction by tree node.
<code>lambda_l1</code>	float	L1 regularization.
<code>lambda_l2</code>	float	L2 regularization.

Name	Type	Description
min_gain_to_split	float	Minimal gain to split a leaf.
drop_rate	float	DART drop rate.
max_drop	int	DART max drop.
skip_drop	float	DART skip drop.
xgboost_dart_mode	bool	DART XGBoost mode.
uniform_drop	bool	DART uniform drop.
top_rate	float	GOSS top retain rate.
other_rate	float	GOSS other retain rate.
min_data_per_group	int	Minimum data per group.
max_cat_threshold	int	Maximum categories thresholds.
cat_l2	float	Categorical L2 regularization.
cat_smooth	float	Categorical smoothing.
max_cat_to_onehot	int	Max categories to one-hot encode.
verbosity	int	Verbosity of the LightGBM backend.
max_bin	int	Max bins.
min_data_in_bin	int	Minimum bin data.
bin_construct_sample_cnt	int	Bin construction sample count.
enable_bundle	bool	Exclusive Feature Bundling (EFB).
zero_as_missing	bool	Use zeroes as missing.
is_unbalance	bool	Unbalanced label.
undersampling	float	The value for undersampling. Only this or oversampling can be used.
oversampling	float	

Name	Type	Description
		The value for oversampling. Only this or undersampling can be used.
seed	int	The seed value.
ignored_fields	set of str	Set of names corresponding to the fields to be ignored when training a model.
schema	Schema	Pass a schema to enable conversion between descs and internal names. By not passing a schema, ds-api will assume field names are references to internal Pulse names, not the ones in the UI.

Examples:

```
>>> model_config = FairGBMConfig(
...     constraint_group_column= "age",
...     bagging_fraction=0.9,
...     undersampling=0.85
... )
>>> model_config.num_leaves # gets default value of num_leaves
>>> model_config.num_leaves = 10 # sets num_leaves to 10
>>> model = model_project.train(
...     "model_name",
...     datasource,
...     "target_field",
...     model_config
... )
>>> model.waitUntilModelTrained()
>>> config_from_model = model.get_config()
```

bagging_fraction property writable [AI](#)

bagging_fraction

bagging_freq property writable [AI](#)

bagging_freq

bin_construct_sample_cnt property writable [AI](#)

bin_construct_sample_cnt

boosting_type property writable [AI](#)

boosting_type

cat_l2 property writable [AI](#)

cat_l2

cat_smooth property writable [AI](#)

cat_smooth

constraint_fnr_threshold property writable [AI](#)

constraint_fnr_threshold

constraint_fpr_threshold property writable [¶](#)

constraint_fpr_threshold

constraint_group_column property writable [¶](#)

constraint_group_column

constraint_stepwise_proxy property writable [¶](#)

constraint_stepwise_proxy

constraint_type property writable [¶](#)

constraint_type

drop_rate property writable [¶](#)

drop_rate

enable_bundle property writable [¶](#)

enable_bundle

feature_fraction property writable [¶](#)

feature_fraction

feature_fraction_bynode property writable [¶](#)

feature_fraction_bynode

global_constraint_type property writable [¶](#)

global_constraint_type

global_target_fnr property writable [¶](#)

global_target_fnr

global_target_fpr property writable [¶](#)

global_target_fpr

ignored_fields property writable [¶](#)

ignored_fields

init_multipliers property writable [¶](#)

init_multipliers

is_unbalance property writable [¶](#)

is_unbalance

lambda_l1 property writable [¶](#)

lambda_l1

lambda_l2 property writable [↗](#)

lambda_l2

learning_rate property writable [↗](#)

learning_rate

lightgbm_seed property writable [↗](#)

lightgbm_seed

max_bin property writable [↗](#)

max_bin

max_cat_threshold property writable [↗](#)

max_cat_threshold

max_cat_to_onehot property writable [↗](#)

max_cat_to_onehot

max_depth property writable [↗](#)

max_depth

max_drop property writable [↗](#)

max_drop

min_data_in_bin property writable [↗](#)

min_data_in_bin

min_data_in_leaf property writable [↗](#)

min_data_in_leaf

min_data_per_group property writable [↗](#)

min_data_per_group

min_gain_to_split property writable [↗](#)

min_gain_to_split

min_sum_hessian_in_leaf property writable [↗](#)

min_sum_hessian_in_leaf

multiplier_learning_rate property writable [↗](#)

multiplier_learning_rate

num_iterations property writable [↗](#)

num_iterations

num_leaves property writable [¶](#)

num_leaves

objective property writable [¶](#)

objective

objective_stepwise_proxy property writable [¶](#)

objective_stepwise_proxy

other_rate property writable [¶](#)

other_rate

oversampling property writable [¶](#)

oversampling

seed property writable [¶](#)

seed

skip_drop property writable [¶](#)

skip_drop

subclasses class-attribute instance-attribute [¶](#)

subclasses: List[[JavaWrapper](#)] = []

top_rate property writable [¶](#)

top_rate

undersampling property writable [¶](#)

undersampling

uniform_drop property writable [¶](#)

uniform_drop

verbosity property writable [¶](#)

verbosity

xgboost_dart_mode property writable [¶](#)

xgboost_dart_mode

zero_as_missing property writable [¶](#)

zero_as_missing

auto_cast staticmethod [¶](#)

auto_cast([config](#): [JavaWrapper](#)) -> [FairGBMConfig](#)

Creates a FairGBMConfig from the java object.

Parameters:

Name	Type	Description	Default
<code>config</code> ⓘ	<code>JavaWrapper</code>	Java configuration object.	<i>required</i>

Returns:

Type	Description
<code>GeneralizedLinearModelConfig</code>	Configuration object according to its python representation.